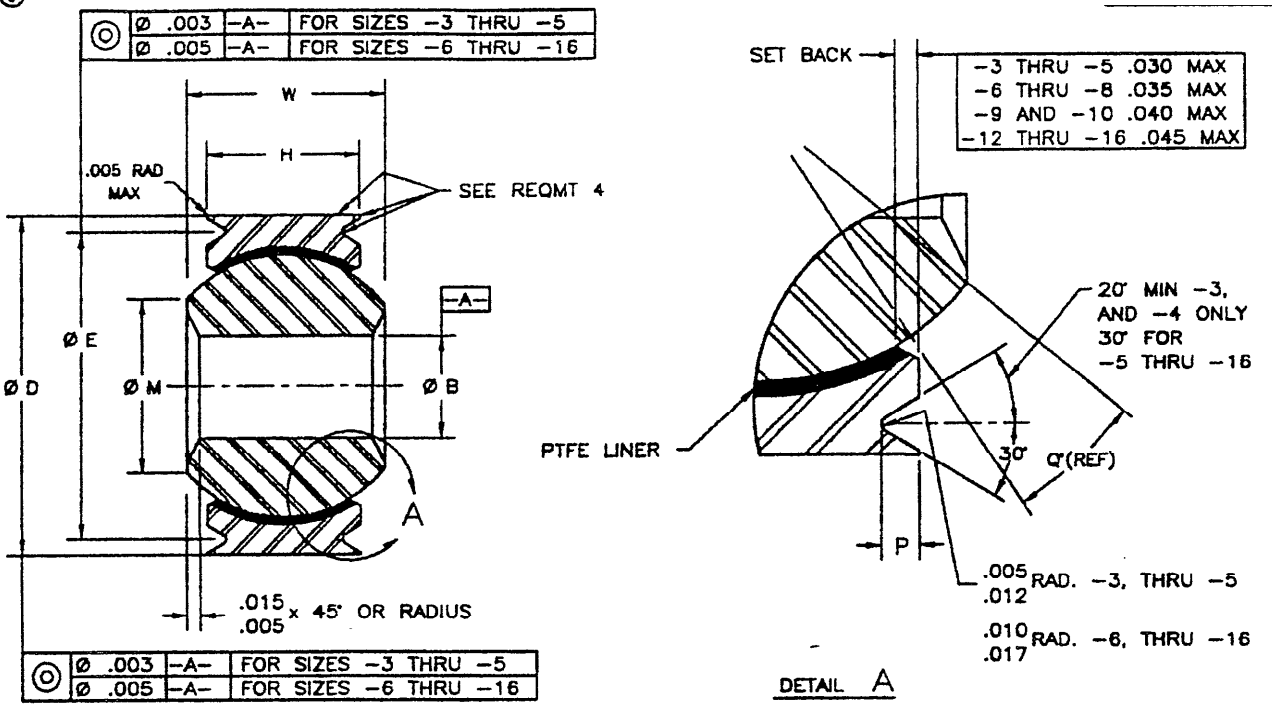
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Ⓙ TABLE I. DIMENSIONS AND STRENGTHS

DASH NO.	ØB +.0000 −.0005	ØD** +.0000 −.0005	H** ±.005	ØM MIN	P +.000 −.010	Q° (REF)	W +.000 −.002	ØE +.000 −.008	STATIC LIMIT LOAD RADIAL LB	STATIC LIMIT LOAD AXIAL LB	OSCILLATING LOAD LB	NO-LOAD ROTATIONAL BREAKAWAY TORQUE IN-LB STANDARD	NO-LOAD ROTATIONAL BREAKAWAY TORQUE IN-LB *"K" TYPE	WT LB MAX (REF)
-3	.1900	.6250	.327	.300	.025	15	.437	.563	2500	1770	4900	0.25-5.0	0-0.5	.031
-4	.2500								5500					
-5	.3125	.6875	.317	.360		14		.625	9400	1640	6050	0.25-8.0	0-1.0	.035
-6	.3750	.8125	.406	.466	.035	8	.500	.712	13700	2630	8310			.060
-7	.4375	.9375	.442	.537		10	.562	.837	20700	3650	11750			.080
-7A	.4375	.9062	.442	.537		10	.562	.806	19700	3650				
-8	.5000	1.0000	.505	.607		9	.625	.900	21400	4970	14950			.100
-9	.5625	1.1250	.536	.721		10	.687	1.025	26600	5370	18100			.135
-10	.6250	1.1875	.567	.747		12	.750	1.087	29000	6130	20250			.160
-12	.7500	1.3750	.630	.845		13	.875	1.251	37000	7730	26200			.240
-14	.8750	1.6250	.755	.995	.055	6	.875	1.501	65200	10800	33600	0.25-12.0	0-2.0	.350
-16	1.0000	2.1250	1.005	1.269		12	1.375	2.001	104000	19300	56250			.970

* SEE REQUIREMENT 5 "NO-LOAD TORQUE" AND NOTE 8.

** SEE NOTE 7.

Ⓙ DENOTES CHANGES

PREPARING ACTIVITY: NAVY - AS
CUSTODIANS: ARMY-AV NAVY-AS
AIR FORCE-99 DLA-
REVIEW: ARMY-AR, AT, ME, DLA-IS, USAF-84
PROJECT NUMBER: 3120-0742-07

MILITARY SPECIFICATION SHEET

TITLE
BEARING, PLAIN, SELF-LUBRICATING, SELF-ALIGNING, LOW SPEED, WIDE, GROOVED OUTER RING, -65° TO 325° F

SPECIFICATION SHEET NUMBER
MS14103 REV J 5 DEC 94

SUPERSEDING MS14103 REV H 24 JAN 94

AMSC - N/A FSC 3120

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(J) TABLE II. OVERSIZE BEARING DIMENSIONS 1/ 2/

RESTRICTED USAGE FOR REPAIR WORK ONLY

.010 AND .020 OVERSIZE OUTSIDE DIAMETER FOR REPLACEMENT OF BEARINGS SHOWN ON SHEET 1

DASH NO.	NOMINAL SIZE	1st OVERSIZE (.010) ØD
-3	.1875	.6350
-4	.2500	.6350
-5	.3125	.6975
-6	.3750	.8225
-7	.4375	.9475
-7A	.4375	.9162
-8	.5000	1.0100
-9	.5625	1.1350
-10	.6250	1.1975
-12	.7500	1.3850
-14	.8750	1.6350
-16	1.0000	2.1350

DASH NO.	NOMINAL SIZE	2nd OVERSIZE (.020) ØD
-3	.1875	.6450
-4	.2500	.6450
-5	.3125	.7075
-6	.3750	.8325
-7	.4375	.9575
-7A	.4375	.9262
-8	.5000	1.0200
-9	.5625	1.1450
-10	.6250	1.2075
-12	.7500	1.3950
-14	.8750	1.6450
-16	1.0000	2.1450

1/ BEFORE INITIATING A REPAIR PROCEDURE TO USE AN OVERSIZE BEARING, APPROVAL FOR MODIFYING AND REIDENTIFYING THE BEARING HOUSING MUST BE OBTAINED FROM THE COGNIZANT ENGINEERING AUTHORITY.

2/ REFER TO NAS0331 FOR installation PROCEDURE AND STAKING FORCES.

REQUIREMENTS:

1. MATERIAL: BALL 440C AMS-5630 OR PH13-8Mo AMS-5629 H1000 (SEE NOTE 8a).
   OUTER RING, AMS-5643 (17-413+).
   UNER PTFE SHALL BE INCLUDED IN THE UNER.
2. SURFACE TEXTURE: BALL DIA Ro B MAX; BORE, BALL, FACE, AND OUTER RING DIA Ro 32 MAX; ALL OTHERS Ro 125 MAX.
3. HARDNESS: BALL, 440C 55-62 Rc OR PH13-8Mo Rc 43 MIN; OUTER RING Rc 28 MIN/Rc 37 MAX BEFORE SWAGING.
4. SURFACE FINISH:

   OUTER RING: PLATING, WHEN SPECIFIED, SHALL BE ZINC-NICKEL PLATING PER AMS-2417, TYPE 2, OR CADMIUM PLATING PER QQ-P-416, TYPE II, CLASS 2, WITH A THICKNESS RANGE OF 0.0003 TO 0.0006 INCHES. PLATE ON THE OUTSIDE DIAMETER SURFACE AND ON THE FLAT BETWEEN THE OUTSIDE DIAMETER AND THE GROOVE. PLATING RUNOUT MAY OCCUR EITHER IN THE GROOVE OR IN THE AREA BETWEEN THE GROOVE AND THE BALL

   BALL: PH13-8Mo, PASSIVATE PER QQ-P-35; 440C, PASSIVATE PER QQ-P-35, OPTIONAL
5. NO-LOAD TORQUE: WHEN THE LETTER " K " IS PRESENT IN THE PART NUMBER, LOWER VALUES OF NO-W TORQUE ARE SPECIFIED PER TAME I IF THE MEASURED TORQUE OF A " K " TYPE BEARING IS LESS THAN 0.1 INCH-POUND, THE INTERNAL RADIAL PLAY SHALL BE MEASURED AND SHALL NOT EXCEED THE FOLLOWING:

DASH NO.	MAXIMUM RADIAL PLAY	MAXIMUM AXIAL PLAY
-3K THRU -12K	0.0007 INCH	0.0021 INCH
-14K THRU -16K	0.0010 INCH	0.0030 INCH

THE INTERNAL AXIAL PLAY SHALL BE MEASURED FOR ALL " K " TYPE BEARINGS.

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PREPARING ACTIVITY NAVY - AS CUSTODIANS: ARMY- AV NAVY- As AIR FORCE-99 DIA- REVIEW ARMY-AR, AT, ME, DIA-IS, USAF-84 PROJECT NUMBER: 3120-0742-07	MILITARY SPECIFICATION SHEET TITLE BEARING, PLAIN, SELF-LUBRICATING, SELF-ALIGNING LOW SPEED WIDE, GROOVED OUTER RING, -65° TO 325° F	SPECIFICATION SHEET NUMBER M S 1 4 1 0 3 REV J 5 DEC 94 SUPERSEDING MS14103 REV H 24 JAN 94 AMSC - N/A FSC 3120

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NOTES:

1. DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE: DECIMALS *.010 AND ANGLES ± 0.5.
2. BREAM SHARP EDGES AND CORNERS AND REMOVE ALL BURRS AND SLIVERS.
3. THE -3 SIZE BEARING IS EXEMPT FROM THE "RADIAL STATIC LIMIT LOAD" TEST BECAUSE THE LOAD CAPACITY OF THE BEARING IS PIN CRITICAL
4. WHEN TESTED TO THE FLUID CONTAMINATION AND SUB-ZERO TEMPERATURE REQUIREMENTS W THE PROCUREMENT SPECIFICATION, THE OSCILLATING LOAD SHALL BE DECREASED TO 75% OF THE SPECIFIED LOAD.
5. WHEN FLUIDS MD ELEVATED TEMPERATURES (ABOVE 200′ F) ARE BOTH PRESENT IN AN APPLICATION, THEN REDUCTIONS IN OPERATING LOADS OR SEARING LIFE MAY BE REQUIRED.
6. DASH NUMBER DESIGNATES NOMINAL BORE DIA IN SIXTEENTHS OF AN INCH.
7. DIMENSION l D" TO BE MET AFTER PLATING. DIMENSION "H " TO BE MET BEFORE PLATING WHEN APPLICABLE.
8. EXAMPLE OF PART NO.

MS14103 -6 C P K T = .375 NOMINAL BORE DIA PH13-8Mo BALL MATERIAL, ZINC-NICKEL PLATED, LOW TORQUE, .010 INCH OVERSIZE D, (DIMENSION 90).

- ABSENCE OF THE LETTERS "T" AND "U " INDICATE STANDARD OUTER DIAMETER (D). (SEE TABLE I).
- T - .010 OVERSIZE D (FOR REPAIR ONLY, SEE TABLE II).
- U = .020 OVERSIZE D (FOR REPAIR ONLY, SEE TABLE II).
- ABSENCE OF THE LETTER "K" INDICATES STANDARD BREAKAWAY TORQUE (SEE TABLE l).
- LETTER "K " INDICATES LOW BREAKAWAY TORQUE (SEE TABLE 1).
- ABSENCE OF THE LETTER "P " INDICATES NO PLATING.
- LETTER "P " INDICATES ZINC-NICKEL OR CADMIUM PLATING (SEE REQUIREMENT 4).
- ABSENCE OF THE LETTER "C " INDICATES 440C BALL MATERIAL.
- LETTER "C" INDICATES PH13-8Mo BALL MATERIAL
- SIZE - BORE DIAMETER (DASH NO. FROM TABLE l).
- MS NUMBER.

80. BALL MATERIAL SHALL BE 440C AMS-5630 UNLESS " C . CALLOUT IS PRESENT IN PART NUMBER. " C" INDICATES PH13-8Mo AMS-5629 H1000.
9. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
10. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS DODISS SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

PREPARING ACTIVITY NAVY - AS CUSTODIANS: ARMY- AV NAVY- As AIR FORCE- 99 DIA- REVIEW: ARMY-AR, AT, ME, DIA-IS, USAF-84 PROJECT NUMBER: 3120-0742-07	MILITARY SPECIFICATION SHEET TITLE BEARING, PLAIN, SELF-LUBRICATING, SELF-ALIGNING LOW SPEED, NARROW, GROOVED OUTER RING, -65" TO 325'F	SPECIFICATION SHEET NUMBER M S 1 4 1 0 3 REV J 5 DEC 91 SUPERSEDING MS14103 REV H 24 JAN 94 AMSC - N/A FSC 3120
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